

Ivonne Berenice Lemus Martínez | Embedded Software Engineer

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Embedded Software Engineer with over 5 years of experience in developing and executing complex embedded software projects. Proficient in **real-time** operating systems (**RTOS**) and **low-level** programming, with a strong knowledge of communication protocols including **CAN**, **I²C**, **SPI**, and **UART**. Demonstrated expertise in utilizing **AUTOSAR** architecture throughout the **automotive** software development lifecycle. Experienced in conducting rigorous testing of secure and critical software systems, ensuring reliability and compliance with **automotive** industry standards.

PROFESSIONAL EXPERIENCE

Embedded Software Engineer - BorgWarner. 19-03-2024 – 09-06-2024

- Integrated **pull request** code updates into the project's main branch using **version control tools** such as **Bitbucket**, **Git**, **GitHub**, **Plastic**, and **SmartGit**.
- Ensured the proper startup of the **bootloader** and application by utilizing **CANoe** and **CANalyzer**, verifying part numbers to confirm correct software deployment.
- Collaborated with a **global cross-functional team** to develop customized solutions that resolved customer needs.

Embedded Software Engineer - CIDE. 16 - 06 2021 - 04-12- 2023

- Improved **ECU** (Electronic Control Unit) reliability by resolving over 20 code **issues** using analysis and **debugging** tools such as **Ozone** and the **JTAG** interface to identify root causes.
- Integrated and configured low-level drivers within the AUTOSAR **Hardware Abstraction Layer (HAL)** to manage communication via **CAN** and **SPI** protocols.
- Configured 10+ signals parameters according with the standard and **Ethernet (TC/IP)** protocols in the **COM** Module of the **Basic Software layer (BSW)**.
- Designed and implemented a **C-based software** component within the **AUTOSAR** architecture to monitor the **ECU** circuit board temperature.
- Implemented software development processes aligned with the **ASPICE V-Model** to support the **Software Development Life Cycle**.
- Optimized software development by 80% through the precise specification of software requirements, in accordance with specified guidelines and following the sprint dynamics of the **Agile methodology**.
- Generated code using **MATLAB/Simulink** and integrated the changes into the software layer in accordance with AUTOSAR interfaces using Vector DaVinci tools.
- Completed tasks on schedule, including software design testing, software requirements analysis, and **peer code reviews**, using **Jira** and **Polarion** management tools.
- Executed hardware testing to ensure detection of short-to-battery, short-to-ground conditions in the ECU's power features, while monitoring **UDS** messages and analyzing system responses.
- Designed software **architecture** using dynamic and activity **UML** diagrams in **Enterprise Architect**, increasing coding efficiency by 80%.
- Provided **mentorship** to engineers in performing low-level testing, unit testing, and static analysis for **safety-critical** systems.
- Used stakeholder tools from **ETAS** and **Vector** to develop software components across **MCAL**, **BSW**, and application layers for wireless charging and **CPD** (Can Power Device) functionality.

- Identified 10+ code vulnerabilities through static analysis using **LDRA** and **TRICORE** tools in Eclipse, reporting **MISRA** violations, **Secure C** issues, and quality warnings.
- Performed software unit tests with VectorCAST to verify the detailed software design, ensuring high-quality software delivery and full condition coverage of functions.

Project Engineer - CIDESI. 12-04-2019 -14-06-2021

- Maximized the efficiency of TMS570LS3137 (**TI microcontroller**) system up to 30%, by developing **embedded software** implementing **RTOS (Real Time Operating System)**, using **GIT** for program version control.
- Developed **low-level drivers** for **I²C** and **UART** communication protocols, supporting **debugging** and validation activities, including **troubleshooting** issues.
- Analyzed over 20 analog schematics of amplifier circuits to perform reverse engineering using an **oscilloscope**.
- Implemented code to read data from digital sensors using **Python** and **embedded Linux**, optimizing GPIO (General Purpose Input/Output) usage by 50%.
- Enhanced the Mexico City railway software by implementing control algorithms in **SafeRTOS** and **FreeRTOS** to manage the speed and acceleration of the railway driving system, resulting in up to a 90% increase in system reliability.
- Developed software in **Embedded C** language for **ARM microcontrollers** to store data in external NAND flash memory, improving storage speed by 80% in a glider datalogger system.
- Implemented **C++ software** using Object-Oriented Programming (**OOP**) to control the speed and acceleration of Mexico City's railway driving system, enabling project flexibility and scalability.
- Visited customer and internal locations to provide hands-on support to **systems engineers**, identify **hardware** requirements, and contribute to the digitalization of the existing railway driving system in Mexico City.
- Improved the Mexico City railway software system by implementing **MISRA-C** and **Secure-C** coding standards, in accordance with ISO 26262 ASIL D **functional safety** requirements.
- Performing **software testing** on train tracks and troubleshooting during system start-up to verify correct implementation of the system update.

Graduate Studies- Instrumented equipment Department - CIDESI. 27-09-2017 – 15-12-2020

- Implemented **I²C** and **SPI** communication protocols on an embedded system using **C language** to transmit and store acquisition data from Hall sensors.
- Implemented a virtual instrumentation array applying least squares method to Model del Dynamic control of a DC control using **MATLAB-SIMULINK** and LabVIEW.

Internship - Instrumentation Equipment Department - CIDESI. 27- 04-2017 - 15-09-2017

B.S. in Mechatronics Engineering | 09 -02-2012 - 10-02-2017

National Technological Institute of Mexico | San Luis Potosí

- Developed a virtual instrumentation applying least squares **algorithms** to model the dynamic control of a DC motor using **Embedded C** language on a **TI microcontroller**.

EDUCATION

B.S. in Mechatronics Engineering | 09 -02-2012 - 10-02-2017

National Technological Institute of Mexico | San Luis Potosí

- Developed **firmware** for Texas Instruments, Microchip, and ST **microcontrollers** with **Assembly** and **C** languages.

M.Sc. & T. Mechatronics | 07-09-2017 - 09-12-2019

CIDESI | Querétaro

-Developed and executed **embedded** software to test a **hardware architecture** for analog data acquisition.

SKILLS

Programing Languages: C, C++ & Assembler

DevOps: Git & Plastic

CPU architectures: FPGA (Artix-7), FPGA (Spartan 3E), TMS570LS3137, MSP430F2618, PIC4550, STL microcontroller.

Communication protocols: I2C, SPI, UART & CAN

Operating systems: RTOS, SafeRTOS & FreeRTOS

Languages: English & Spanish

COURSES

C++: UDEMY. 2024-09-10 - 2024-10-10

Vector tools: CIDEDEC. 2021-06-16 - 2021-09-27